A determinant certificate for canonical Bell inequalities

23 October 2026

A canonical Bell inequality separates into eight positive event terms and one negative state component. This structure reduces positivity of its four-dimensional state matrix to an angular determinant. That determinant has only eight nonconstant monomials, all with nonnegative coefficients. Their exponents form twelve elementary balanced circuits, each with an explicit sum-of-squares identity.

We combine these two observations into a single certificate for every canonical valid inequality and every measurement angle. Multiplication by the determinant of the three-dimensional positive state block clears the state elimination, including its singular limits. The resulting certificate uses 104 fixed functions and an explicit rational semidefinite pencil of order 133. Its coefficients depend quartically on the inequality. This dependence is the remaining step between the certificate and an affine semidefinite representation of the quantum set.

1 Completing the state quadratic without division

Use the compact half-angle cover

$$a_0^2 + a_1^2 = b_0^2 + b_1^2 = 1, \qquad \psi = (W, X, Y, Z) \in S^3.$$

Thus $A_0 = B_0 = \sigma_z$, $A_1 = (a_0^2 - a_1^2)\sigma_z + 2a_0a_1\sigma_x$, and similarly for Bob. The circuit identities below contain cube roots of angular monomials. To make those roots polynomial on a fixed cover, adjoin all real solutions of

$$x_0^6 = a_0^2, \quad x_1^6 = a_1^2, \quad y_0^6 = b_0^2, \quad y_1^6 = b_1^2. \tag{1}$$

This compact algebraic cover includes every root sign, angle sign and projective endpoint. Every new variable has absolute value at most one.

For the canonical source use $d \geq 0$ and eight nonnegative event weights $(\lambda, n, m, C, A, D, \alpha, \beta)$, with

$$\kappa = (4\lambda, 4n, 4m, 4C, 4A, 4D + 2d, 4\alpha + 2d, 4\beta + 2d).$$

The eight rows below contain no source coefficients:

i	R_{iW}	R_{iX}	R_{iY}	R_{iZ}
1	b_0	b_1	0	0
2	0	1	0	0
3	a_0b_1	$-a_0b_0$	a_1b_1	$-a_1b_0$
4	a_1	0	$-a_0$	0
5	0	0	1	0
6	a_1b_0	a_1b_1	$-a_0b_0$	$-a_0b_1$
7	0	0	b_1	$-b_0$
8	0	a_0	0	a_1

(2)

The complete canonical polynomial, including the signed core, is

$$F = \sum_{i=1}^{8} \kappa_i (R_i\psi)^2 - 2dZ^2 = \psi^T P \psi, \qquad P = \sum_i \kappa_i R_i^T R_i - 2de_4e_4^T. \tag{3}$$

These are precisely the homogeneous physical-parent rows, restricted to the normalized half-angle cover. Put $r_i = (R_{iW}, R_{iX}, R_{iY})^T$ and write $P = \left[\begin{smallmatrix} H & b \\ b^T & c \end{smallmatrix}\right]$. In particular $H = \sum_i \kappa_i r_i r_i^T$.

The adjugate of H is a sum of outer products of cross products of its rows. Completing the state quadratic therefore produces the functions

$$f_{ijk} = \det(r_i, r_j, r_k) R_k \psi, \qquad 1 \le i < j \le 8, \quad k \notin \{i, j\}. \tag{4}$$

The 168 indices include zero functions and linear dependencies. Let $E = \{-1, 0, 1\}^2 \setminus \{0\}$ and

$$\mathcal{N} = 3E \ \cup \ \{(\pm 1, \pm 1), (\pm 1, \pm 2), (\pm 2, \pm 1)\} \ \cup \{0\}. \tag{5}$$

There are exactly 21 distinct elements. For $\nu \in \mathcal{N}$ define

$$g_\nu = 4 x_0^{3-\nu_1} x_1^{3+\nu_1} y_0^{3-\nu_2} y_1^{3+\nu_2} Z. \tag{6}$$

Every exponent in (6) is an integer between zero and six.

Theorem 1 (Fixed multiplier certificate)**.** *For every quantum-valid canonical source with $d \ge 0$ and all eight event weights nonnegative, the polynomial $(\det H)F$, on the entire testing cover (1), belongs to the Gram cone of the fixed indexed family*

$$(f_{ijk}) \ \cup \ (g_\nu), \tag{7}$$

of size 189. A positive-semidefinite source-only Gram matrix is specified in (9), (15)–(16) below. If the angular determinant has a positive stationary zero, g_0 is unnecessary, giving the 188-function subfamily. No source weight is required to be positive, and no testing opening is required to be nonzero.

The matrix completion is the following polynomial identity, valid even when H is singular:

$$(\det H)F = \sum_{i<j} \kappa_i \kappa_j \left(\sum_{k \notin \{i,j\}} \kappa_k f_{ijk} \right)^2 + (\det P) Z^2. \tag{8}$$

Indeed Cauchy–Binet gives $\operatorname{adj} H = \sum_{i<j} \kappa_i \kappa_j (r_i \times r_j)(r_i \times r_j)^T$. Also $[H \ b]\psi = \sum_k \kappa_k r_k (R_k \psi)$, and the block determinant identity gives $\det P = c \det H - b^T (\operatorname{adj} H) b$. Substituting these three polynomial identities proves (8); neither an inverse nor a division has been used. Thus the first Gram block, in the ordering (4), is the explicit block diagonal matrix

$$G_{\text{state}} = \bigoplus_{i<j} \kappa_i \kappa_j \, \kappa_{\widehat{ij}} \kappa_{\widehat{ij}}^T, \qquad \kappa_{\widehat{ij}} = (\kappa_k)_{k \notin \{i,j\}}. \tag{9}$$

2 A fixed angular Gram cone for every valid source

It remains to express the determinant term in (8) as fixed squares. Introduce affine variables $u = a_1/a_0$, $v = b_1/b_0$, $U = u^2$, $V = v^2$. Write $D_u = 1 + u^2$, $D_v = 1 + v^2$. The affine state matrix obeys

$$\det P_{\text{aff}} = 16 D_u^2 D_v^2 u^2 v^2 L(U, V), \qquad L = h + \sum_{e \in E} q_e U^{e_1} V^{e_2}, \quad q_e \ge 0. \tag{10}$$

For reproducibility, set

$$\begin{aligned} t_1 &= d + 2(\lambda + n + D), & t_2 &= d + 2(\alpha + m + A), \\ t_3 &= d + 2(\lambda + C + D), & t_4 &= d + 2(\alpha + \beta + m), \\ t_5 &= d + 2(\lambda + \beta + n), & t_6 &= d + 2(\alpha + C + A), \\ t_7 &= d + 2(\beta + n + m), & t_8 &= d + 2(C + A + D). \end{aligned}$$

The coefficient indexed by $e = (i-1, j-1)$ is

$$\begin{aligned} q_{00} &= 4\lambda\alpha t_7 t_8, & q_{01} &= 4(\lambda D t_7 t_6 + \alpha m t_5 t_8), \\ q_{02} &= 4mD t_5 t_6, & q_{10} &= 4(\alpha A t_7 t_3 + \lambda n t_8 t_4), \\ q_{12} &= 4(\beta m t_1 t_6 + CD t_5 t_2), & q_{20} &= 4nA t_3 t_4, \\ q_{21} &= 4(\beta n t_3 t_2 + CA t_1 t_4), & q_{22} &= 4\beta C t_1 t_2, \end{aligned}$$

and $h = \det P_{\mathrm{aff}}(1,1)/256 - \sum_e q_e$. Expanding the determinant of (3) gives these coefficients. Validity of the Bell inequality implies $L(U,V) \geq 0$ for all $U, V > 0$.

Lemma 1 (The 21-monomial Laurent cone)**.** *Every $L = h + \sum_{e \in E} q_e U^{e_1} V^{e_2}$ with $q_e \geq 0$ and $L \geq 0$ on the positive quadrant is a sum of squares in the fixed span*

$$\{U^{\nu_1/6} V^{\nu_2/6} : \nu \in \mathcal{N}\}. \tag{11}$$

The source-only algebraic coefficient prescription below also covers zero outer coefficients and minima attained only at infinity.

Proof. If $h \geq 0$, use $\sqrt{h}$ and $\sqrt{q_e} U^{e_1/2} V^{e_2/2}$. Suppose $h < 0$ and put $I = \{e : q_e > 0\}$. The origin belongs to $\operatorname{conv} I$: otherwise strict separation and a ray in logarithmic coordinates make every nonconstant term tend to zero, contradicting $h < 0$. Let J consist of the exponents on the minimal face of $\operatorname{conv} I$ containing the origin. An exposing ray sends every off-face monomial to zero while keeping the face monomials fixed. Consequently

$$h + \sum_{e \in J} q_e U^{e_1} V^{e_2} \geq 0. \tag{12}$$

In the span of J, the function $\sum_{e \in J} q_e \exp(e \cdot y)$ is coercive and strictly convex: the origin lies in the relative interior of $\operatorname{conv} J$, and its Hessian is $\sum q_e \exp(e \cdot y) e e^T$, positive on that span. It therefore has a unique minimum modulo the orthogonal complement. Choose a positive minimizer (U_c, V_c) and put

$$w_e = q_e U_c^{e_1} V_c^{e_2} \ (e \in J), \quad w_e = 0 \ (e \notin J), \qquad c = \sum_{e \in J} w_e, \qquad r = h + c \geq 0.$$

Then $\sum w_e e = 0$. For a rank-two face, (U_c, V_c) is the unique positive solution of the two equations

$$\sum_{e \in J} e_1 q_e U_c^{e_1+1} V_c^{e_2+1} = 0, \qquad \sum_{e \in J} e_2 q_e U_c^{e_1+1} V_c^{e_2+1} = 0. \tag{13}$$

These have bidegree at most $(2,2)$ and prescribe algebraic functions of the source coefficients. For a rank-one face $J = \{e, -e\}$ choose e with first nonzero component positive, set $t = \sqrt{q_{-e}/q_e}$, and take $(U_c, V_c) = (t, 1)$ if $e_1 = 1$, or $(1, t)$ if $e = (0, 1)$. This gives $U_c^{e_1} V_c^{e_2} = t$.

The face selection needs only fixed rational sign cases. Order the four directions as $(1,0)$, $(0,1)$, $(1,1)$, $(1,-1)$, and for each direction try $\sigma = 1$ before $\sigma = -1$. Test whether $q_e, q_{-e} > 0$ and $q_f = 0$ for every f with $\sigma \det(e, f) < 0$. The first successful test gives $J = \{e, -e\}$. If none succeeds, $J = I$ is the rank-two case. These are exhaustive because a proper face through the origin must be one of the four opposite pairs in E.

The nonnegative balanced vector w is a conic combination of exactly twelve primitive positive circuits: four opposite pairs with weights $(1,1)$, four triangles $\{(s,t), (-s,0), (0,-t)\}$ with weights $(1,1,1)$, and four triangles formed by adjacent corners and the opposite axial point, with weights $(1,1,2)$. Here $s, t \in \{-1, 1\}$. A deterministic prescription chooses the first supported circuit in the fixed order in the accompanying JSON file, subtracts $\lambda_C b^C$ with $\lambda_C = \min_{b_e^C > 0} w_e / b_e^C$, and repeats. Each subtraction eliminates a positive entry, so at most eight steps are needed.

Put $p_e = (U/U_c)^{e_1} (V/V_c)^{e_2}$. An opposite pair contributes $(p_e^{1/2} - p_{-e}^{1/2})^2$. For a triangle of weight $(1,1,1)$ use the nine squares

$$\frac{1}{2} \sum_{h \in C} \sum_{\{l,m\} \subset C} \left[p_h^{1/6} (p_l^{1/3} - p_m^{1/3}) \right]^2 = \sum_{e \in C} p_e - 3. \tag{14}$$

For weights $(1,1,2)$, with $e+f+2g=0$, use $(p_e^{1/2}-p_f^{1/2})^2+2(p_{-g}^{1/2}-p_g^{1/2})^2=p_e+p_f+2p_g-4$. All exponents in these roots belong to (11): in (14) they are $(h+2l)/6$, and the possible numerators are precisely among (5). Multiply each circuit identity by λ_C, and add the constant r and the off-face monomials. Their sum is exactly L. □

For an entirely explicit Gram prescription, let e_ν denote the coordinate vector indexed by $\nu\in\mathcal{N}$. The twelve fixed rational positive-semidefinite matrices are

$$\begin{aligned}
&A_C=(e_{3e}-e_{-3e})(e_{3e}-e_{-3e})^T, && C=\{e,-e\},\\
&A_C=\tfrac{1}{2}\sum_{h\in C}\sum_{\{l,m\}\subset C}(e_{h+2l}-e_{h+2m})(e_{h+2l}-e_{h+2m})^T, && C \text{ has weights } (1,1,1),\\
&A_C=(e_{3e}-e_{3f})(e_{3e}-e_{3f})^T+2(e_{-3g}-e_{3g})(e_{-3g}-e_{3g})^T, && e+f+2g=0.
\end{aligned} \tag{15}$$

Their entries are evaluated in `multiplier_schema.json`. Set $T=\operatorname{diag}(U_c^{-\nu_1/6}V_c^{-\nu_2/6})$. The angular Gram matrix is

$$G_{\mathrm{ang}}=\begin{cases} he_0e_0^T+\sum_{e\in E}q_e e_{3e}e_{3e}^T, & h\ge 0,\\ re_0e_0^T+\sum_{e\notin J}q_e e_{3e}e_{3e}^T+T(\sum_C\lambda_C A_C)T, & h<0. \end{cases} \tag{16}$$

It is positive semidefinite and depends only on the source. Its coefficients use finite support and sign cases, positive algebraic roots, the unique positive solution (13), and finitely many minima. There is no testing-dependent coefficient choice.

On $a_0a_1b_0b_1\neq 0$ the homogeneous form of (10) is

$$\det P=16a_0^2a_1^2b_0^2b_1^2L(a_1^2/a_0^2,b_1^2/b_0^2).$$

On the nonnegative-root branch, multiplication of (11) by $4|a_0a_1b_0b_1|Z$ gives exactly (6), so $\det P\,Z^2=g^TG_{\mathrm{ang}}g$ there. The root-sign and endpoint statements are algebraic. In every dyad in (15), the two indices are congruent coordinatewise modulo two. Consequently every nonzero entry of G_{ang} joins indices with the same parities. Each product $g_\nu g_\mu$ appearing in the quadratic form therefore has even exponents in all four new variables. The quadratic identity is unchanged by any of their sign flips. In each fixed Laurent identity set $U=(x_1/x_0)^6$, $V=(y_1/y_0)^6$ and multiply by $16x_0^6x_1^6y_0^6y_1^6Z^2$. Every exponent is nonnegative, because each index lies in $[-3,3]^2$. This gives a polynomial identity also at zero coordinates, and $x_i^6=a_i^2$, $y_i^6=b_i^2$ identify its right-hand side with the homogeneous determinant. Thus every zero opening and projective endpoint is included without division. Together with (8)–(9) this proves the theorem with Gram matrix $G_{\mathrm{state}}\oplus G_{\mathrm{ang}}$. For a positive stationary zero of L, take its contact as (U_c,V_c), $J=I$, and $r=0$; the constant angular feature is then absent.

Independent real closed fields. The Laurent lemma, including the finite support split and the prescriptions (13)–(16), is first-order over the reals after its source-side positive roots are represented by polynomial equations and sign conditions. It therefore holds over any real closed source field K_c extending $\mathbb{R}$. The Gram coefficients and the square roots of the nonnegative source quantities used in their dyad factorizations belong to K_c. Equations (8), (14), and the polynomial homogenization just given then hold in $K_c\otimes_{\mathbb{R}}K_t$ for an independent real closed testing extension K_t of $\mathbb{R}$. The resulting tensor roots use exactly the fixed testing functions (7); no source-dependent testing radical or bounded-residue choice occurs.

Corollary 1 (Exact quartic pullback, with both inclusions)**.** *Among canonical tuples with d and all eight event weights nonnegative, physical Bell validity is equivalent to membership of $(\det H)F$ in the fixed Gram cone (7). Consequently the canonical valid-slack cone is the inverse image of a fixed rational Gram cone under an explicit homogeneous quartic coefficient map.*

Proof. The theorem proves the forward implication. Conversely, a Gram representation makes $(\det H)F$ nonnegative at every test. If $d > 0$, the signed-core part of the affine $Z = 0$ block is

$$2d\{(uW + uvX - Y)^2 + (1 + u^2)v^2Y^2 + (1 + v^2)X^2\}.$$

It is strictly positive for every nonzero (W, X, Y) when $uv \neq 0$; hence $\det H > 0$ on a dense open set of tests. There $F \geq 0$, and continuity extends the inequality to every projective test. If $d = 0$, the original parent (3) is already a sum of nonnegative event squares. The Jordan block reduction for two binary measurements, proved in Chapter 2 of the companion, reduces general tests to two-dimensional blocks. Local basis choices place each pair in the real plane used here. Since P is real, its value on a complex state is the sum of its values on the real and imaginary parts. Thus the displayed real-state tests establish validity on the full quantum set. Finally H and F are linear in the source, so $(\det H)F$ has source degree four. Every testing function in (7) is a polynomial with rational coefficients on (1), so all Gram product equations are rational and finite. □

3 The rational pencil

All data in this section are evaluated over $\mathbb{Q}$ in the accompanying JSON files. Reduce polynomial feature products by the six monic rules

$$\begin{gathered} a_0^2 = 1 - x_1^6, \quad a_1^2 = x_1^6, \quad b_0^2 = 1 - y_1^6, \quad b_1^2 = y_1^6, \\ x_0^6 = 1 - x_1^6, \qquad y_0^6 = 1 - y_1^6. \end{gathered} \tag{17}$$

Their leading monomials are pairwise coprime, so they form a monic Gröbner basis for the angle-cover equations. Every feature product is homogeneous quadratic in the state; its vanishing on S^3 is equivalent to its vanishing on every state. No state-sphere reduction is needed. These normal forms introduce no spurious functional identities: the real sign sheets over the open arcs of the two Fermat curves are Zariski dense. Equivalently, sign averaging separates the four half-angle parities, and the remaining reduced polynomials vanish only when all their coefficients vanish on the two curves.

The 189 indexed functions have 175 distinct normal-form monomials and coefficient rank 104: 83 polynomial functions and 21 angular functions. The exported 104-column basis and its exact rational reconstruction matrix express every indexed function in that basis. In the 28 state blocks of (9), two blocks vanish, 22 have function rank four, and four have rank six. Congruence by their exported rational basis maps gives the fixed structured Gram cone with block orders

$$\underbrace{4, \ldots, 4}_{22}, \quad \underbrace{6, \ldots, 6}_{4}, \quad 21. \tag{18}$$

Its total order is 133 and it has 535 symmetric Gram coordinates. The exact source blocks are pushed forward by the same congruences; positivity and every represented polynomial are preserved.

Let $z \in \mathbb{R}^{535}$ be those upper-triangular Gram coordinates. With off-diagonal products counted twice, the evaluated product matrix is

$$\mathcal{A} \in \mathbb{Z}^{261 \times 535}, \qquad \operatorname{rank} \mathcal{A} = 189. \tag{19}$$

It has 5653 nonzero entries. Choose the 189 pivot rows J and pivot columns P recorded in the export, and let K denote the other 346 columns. Put $B = \mathcal{A}_{J,P}$. All coefficient equations are eliminated exactly by

$$z_K = t, \qquad z_P = B^{-1}y - B^{-1}\mathcal{A}_{J,K}t. \tag{20}$$

Inserting (20) into the block matrices (18) gives an actual rational pencil $\mathcal{L}(y, t)$ of order 133, with 189 visible coefficient variables, 346 free Gram variables, and 6079 nonzero upper-triangle

coefficient records. The export includes the coefficient reconstruction $\mathcal{A}_{:,P}B^{-1}y$, so all 261 product equations are specified, including their 72 linear dependencies. Conversely, every feasible Gram tuple is represented by (20), because its K coordinates are free and B is invertible. Thus this pencil describes the structured Gram cone exactly, in both directions.

For a canonical source $c = (d, \lambda, n, m, C, A, D, \alpha, \beta)$, let

$$y(c) = \operatorname{coeff}_J\big((\det H_c)F_c\big). \tag{21}$$

The evaluated map (21) has 7472 rational monomial records, all of homogeneous source degree four. It is obtained from (9), with the angular target initially represented by the diagonal entries h, q_e; this initial diagonal representation is an identity, not a claim that h is nonnegative. The circuit construction supplies a positive Gram when the source is valid. The preceding theorem and its reverse implication give the exact statement

$$\boxed{c \geq 0: \quad F_c \text{ is quantum-valid} \iff \exists t \in \mathbb{R}^{346}\ \mathcal{L}(y(c), t) \succeq 0.} \tag{22}$$

For a fixed source, (22) is finite semidefinite feasibility. The input $y(c)$ is quartic. Treating its monomials as free lift variables would drop their defining relations and would not preserve (22). In particular, the 189 coefficient variables are not the eight quantum expectations. Formula (22) is an exact polynomial pullback of a spectrahedral shadow, not an affine lift of Q_{222}.

4 Removing the determinant multiplier

The source-dependent multiplier in (8) cannot simply be removed by a fixed finite family of exact inverse or whitening factors, even on compact regions with uniform positive reserve. Here is a physical example.

Take $d = 1$, $\lambda = n = m = C = \alpha = \beta = 1$, $D = 2$, and $A = a \in [1, 2]$. For the CHSH convention $S = E_{00} - E_{01} + E_{10} + E_{11}$, the sum of the eight losing event probabilities is $2 - S/2$. Hence this Bell slack is

$$f_a = 9 - \frac{5}{2}S + 4(a-1)p_A + 4p_D \geq 9 - 5\sqrt{2} > \frac{3}{2}. \tag{23}$$

The inequality follows directly from $S^2 = 4I + [A_0, A_1] \otimes [B_0, B_1]$ and $\|[A_0, A_1]\|, \|[B_0, B_1]\| \leq 2$. Thus every source in this compact family is strictly quantum valid.

At the noncommuting affine tests $u = 1$ and $v \in [1/2, 1]$, its $Z = 0$ block is exactly

$$H_a(v) = \begin{pmatrix} 8v^2 + 22 & 14v & -14 \\ 14v & 32v^2 + 18 & -14v \\ -14 & -14v & 8a(1+v^2) + 20v^2 + 14 \end{pmatrix}. \tag{24}$$

In particular $\frac{15}{4}I \preceq H_a(v) \preceq 94I$ throughout this compact source–test rectangle. The lower bound follows from (23) and $F = 2(1+v^2)\|\psi\|^2 f$; the upper bound follows from the absolute row sums of (24). Direct evaluation gives

$$(H_a(v)^{-1})_{YY} = \frac{1}{8(1+v^2)(a + r(v))}, \qquad r(v) = \frac{320v^4 + 621v^2 + 126}{2(1+v^2)(64v^2 + 99)}. \tag{25}$$

The rational function r is nonconstant; for example, $r(1/2) = 241/230$ and $r(1) = 1067/652$.

Proposition 1 (Rank-one inverse obstruction)**.** *For any distinct $a_1, \ldots, a_N \in [1, 2]$, the functions $(H_{a_j}^{-1})_{YY}$ are linearly independent on $[1/2, 1]$. Consequently no finite-dimensional real space of testing functions can provide exact inverse Gram factors $H_a^{-1} = B_a B_a^T$ for every $a \in [1, 2]$. In particular no such space supports square invertible exact whitening matrices $C_a^T H_a C_a = I$ for this family.*

Proof. After multiplication by $8(1+v^2)$, a dependence would have the form $\sum_j c_j/(a_j + r(v)) = 0$. Multiplying denominators produces a polynomial $Q(r(v)) = 0$. Since a nonconstant rational function takes infinitely many values on an interval, $Q = 0$. Evaluating $Q(-a_j)$ then gives $c_j \prod_{k\neq j}(a_k - a_j) = 0$, so every $c_j = 0$. If all entries of all B_a belonged to a fixed finite space V, every entry of $B_a B_a^T$ would belong to the finite-dimensional product span $\operatorname{span}\{vw : v, w \in V\}$, contradicting this independence. Finally $C_a^T H_a C_a = I$ with C_a square invertible implies $H_a^{-1} = C_a C_a^T$. □

This proposition rules out that exact-whitening construction. It does not rule out other fixed Gram representations of the original slack, and it is not an obstruction to the existence of a finite lift of the quantum set. In fact, the example itself has a fixed unmultiplied certificate: with $T_0 = A_0 - (B_0 - B_1)/\sqrt{2}$ and $T_1 = A_1 - (B_0 + B_1)/\sqrt{2}$, the identity $T_0^2 + T_1^2 = \sqrt{2}(2\sqrt{2}I - S)$ gives

$$9I - \tfrac{5}{2}S = (9 - 5\sqrt{2})I + \tfrac{5}{2\sqrt{2}}(T_0^2 + T_1^2).$$

Its two additional event terms are already positive squares. The precise unfinished step is to obtain a source-independent finite Gram cone for F itself from an identity such as (8), without dividing by the mixed source–testing function $\det H$ or introducing source-dependent testing functions. At some zero-weight boundary sources the multiplier also vanishes at commuting tests; the continuous identity proved above does not turn that zero into an unmultiplied certificate.